MARY · 09:02 · TALLAGHT ED



«Допоможіть, будь ласка. У мене дуже болить у грудях. У мене алергія на пеніцилін.»

Asking for help is the most basic thing a person in need can do.

But what happens when nobody understands you?

CareBridge

No patient goes unheard · Bakht Zia Khan · Chijindu Okafor · Eriitunu Adesioye · Mohd Wafa

Mary is a fictional patient, built from the real figures on page 2. Her line is the Ukrainian voice clip we play in our live demo; page 3 shows what CareBridge heard.

Ukrainian is TUH's most-booked interpreter language (IJMS study, reported 21 Apr 2026): https://www.waterford-news.ie/dublin-hospital-hired-interpreters-for-foreign-language-patients-nearly-11000-times-in-17-months_arid-95325.html

Mary's emergency

Mary is visiting her daughter in Dublin. Her chest starts to hurt. She is in pain, she is scared, and she doesn't speak English.

09:02 ● Mary arrives at Tallaght and tries to explain what's wrong.

09:04 ● The nurse opens a translation app. First, it wants a language.

TRANSLATE FROM	
Russian	?
Polish	?
Ukrainian	?
Romanian	?

09:12 ● **Still guessing.** Nobody knows she is allergic to penicillin. In an emergency, minutes matter.

Mary is not the problem. The system is.

10,793
times Tallaght needed an interpreter in 17 months[1]

Top 4
Ukrainian, Russian, Polish, Romanian: the very languages the nurse was guessing between[1]

1 in 12
machine-translated hospital sentences that could cause significant harm[2]

1 TUH interpreter-use study, Jan 2024 to Jun 2025, Irish Journal of Medical Science (reported 21 Apr 2026): https://www.waterford-news.ie/dublin-hospital-hired-interpreters-for-foreign-language-patients-nearly-11000-times-in-17-months_arid-95325.html

2 Khoong et al., JAMA Internal Medicine, 2019. Google Translate on emergency discharge instructions: up to 8% of sentences in Chinese, 2% in Spanish: https://pubmed.ncbi.nlm.nih.gov/30801626/

The timeline is an illustration.

One tap.
Mary is heard.



Real output from CareBridge on 10 Oct 2026, from Mary's voice clip. Read out loud in seconds.

1 Tap once

One big button: **START NOW**. No menus, no language to choose.

2 Mary speaks

CareBridge works out that she speaks Ukrainian, not Russian, and locks it in for the whole visit.

3 Everyone hears

Her words are spoken in English for the nurse. The nurse's words are spoken in Ukrainian for Mary. Out loud, both ways.

THE NURSE ANSWERS

"You are safe. We will not give you penicillin."

Ти в безпеці. Ми не дамо вам пеніциліну.

✓ "Not" kept ✓ Medicine kept

No menus. No guessing. Put it on the table and talk.

For the first time since she walked in, Mary can say what's wrong.

Screen redrawn from CareBridge's Emergency mode with the live output of 10 Oct 2026 (patient line translated and checked in 3.9 s, reply in 2.6 s, after speech recognition).
Speech: OpenAI Whisper https://arxiv.org/abs/2212.04356 · Translation: Meta NLLB-200 https://arxiv.org/abs/2207.04672 · Voices: Piper https://github.com/rhasspy/piper

Safe enough for a hospital. Private enough for Mary.

Checked before anyone hears it

- Every number
- Every “not”
- Left and right
- Every medicine

27 / 28

We planted 28 dangerous errors in real translations: wrong doses, missing “nots”, swapped sides and medicines. CareBridge caught 27, with **zero false alarms**.

Consent? Bad news? The AI stops, and one tap calls a human interpreter. Some conversations should always have a person.

Saved to Mary’s record

Mary · Ukrainian · Emergency ✓ SAVED

Patient · 09:04
Help me, please. My chest hurts a lot, I’m allergic to penicillin.

Nurse · 09:04
You are safe. We will not give you penicillin.

Allergy: penicillin

Her care team can come back to it
What she said and how it was translated, as a standard health record.

Her voice is never saved
The audio is processed in memory and thrown away. Only the words are kept.

Her data stays in Ireland
In the hospital, CareBridge runs on AWS in Dublin. Staff sign-in only.

Tested before it’s trusted

37 → 12
languages tested on 20 medical sentences each. 28 passed; 12 are in our pilot, led by TUH’s top four.

Failed = off
Mandarin turned “appendicitis” into “tachycardia”, so it was switched off, not shipped.

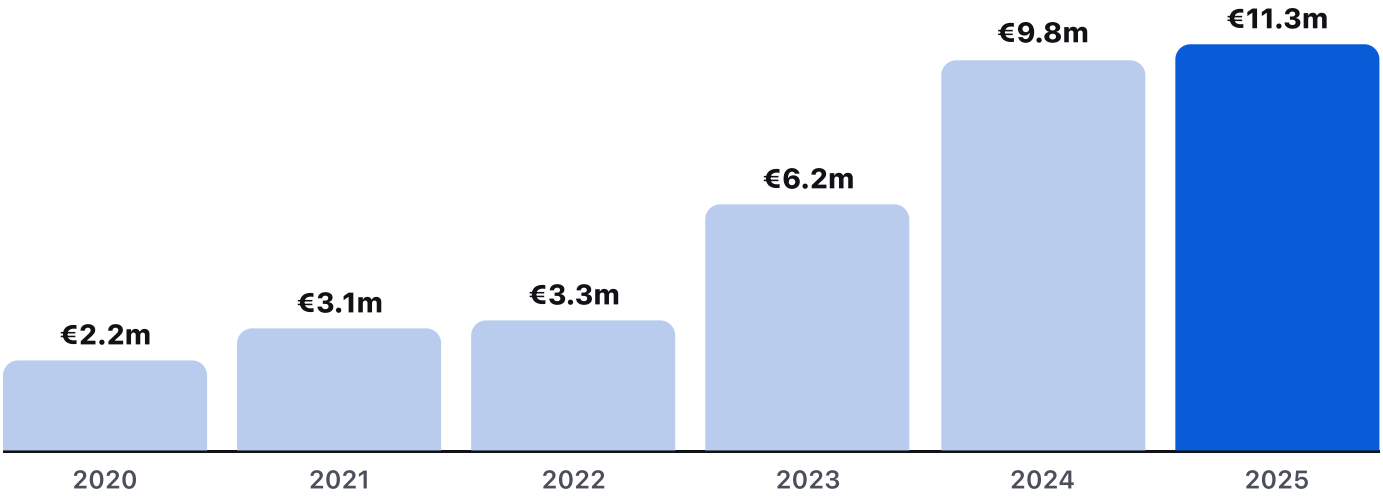
Mary is told
On her screen, in her own language, that a computer is translating, and that she can ask for a person at any time.

Stress test and language tests run on the live system, 9 to 10 Oct 2026 (CareBridge repository: eval/error_injection.py, eval/results.json, eval/lang_eval.json). Record card redrawn from the CareBridge audit log. Record format: HL7 FHIR DocumentReference https://hl7.org/fhir/R4/documentreference.html

AWS Europe (Ireland) region, eu-west-1: https://aws.amazon.com/about-aws/global-infrastructure/regions_az/ · Our hackathon demo runs in the AWS lab region we were given; Dublin is the hospital set-up.

€11.3 million

spent by the HSE on interpreting and translation in 2025. **Five times what it spent in 2020.**[1]



€530,331

Letterkenny University Hospital alone, in 2025, across 39 languages.[2]

Under €3

a day to run the server behind our live demo, around the clock.[3]

Interpreters are essential. But they can't be at every bedside at 3am.

24/7	Every room	First second
day and night	any phone, iPad or laptop	no booking, no waiting

1 HSE reply to PQ 31617/26 (Deputy Carol Nolan), 7 May 2026. HSE spend on translation and interpreter services: €2,244,752 in 2020, €11,287,959 in 2025: https://about.hse.ie/publications/question-from-deputy-carol-nolan-pq-31617-26/

2 Donegal News, HSE FOI release on interpreting spend at Letterkenny University Hospital, 2025, incl. VAT: https://donegalnews.com/revealed-hse-spend-on-interpreters-at-luh

3 One Amazon EC2 m7i.large server, on demand, about US$0.10 to $0.11 an hour: https://aws.amazon.com/ec2/pricing/on-demand/

CareBridge

From the emergency room to the follow-up call.

IN THE APP TODAY

Consultation Mode

For Mary's check-ups, when there is time to get every word right.

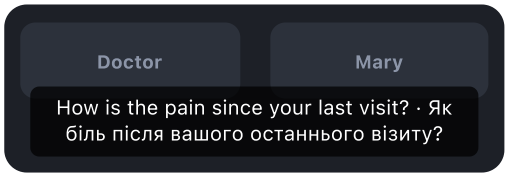
1. **The doctor speaks**
   Into their phone, in plain English.
2. **The doctor checks**
   CareBridge shows what it heard and highlights every dose and medicine. The doctor fixes anything wrong and confirms.
3. **Mary hears it**
   On her own screen, read aloud in Ukrainian. She answers by voice or with a tap.
4. **It's on her record**
   Every word, in both languages, ready for her next visit.

> DOCTOR SAID Take two tablets of 500 milligram paracetamol every six hours.
> MARY HEARS Прийміть дві таблетки 500 мг парацетамолу кожні шість годин.
> ✓ Numbers kept: 2 · 500 · 6 ✓ Medicine kept

NEXT STEP

Video follow-ups

When Mary goes home, her care comes with her: CareBridge inside Google Meet, with live translated subtitles for Mary and her doctor.



CareBridge doesn't replace the people who care for Mary.
It makes sure they can understand her.

Because everyone deserves to be able to ask for help, and to be understood when they do.

Scan to try CareBridge live

Staff sign-in only. Ask any of us for today's demo code.

Bakht Zia Khan · Chijindu Okafor · Eriitunu Adesioye · Mohd Wafa
TUH Healthcare Innovation Challenge 2026 · Challenge 6: real-time medical multilingual speech translation

Consultation Mode example: live CareBridge server, 10 Oct 2026 (four details highlighted; numbers and medicine checks passed). The video-call picture is a sketch of our next step; it is not built yet.
Google Meet is a product of Google LLC; CareBridge is not affiliated with Google. Live app: https://lekjx2h7ch.execute-api.us-east-1.amazonaws.com